

$$* T(n) = T(n/2) + T(n/2) + n$$

Left half Right half

Merge of 2 parts

$T(1) \longrightarrow$ if $n=1$

$$T(n) = \begin{cases} 2T(n/2) + n & \text{if } n > 1 \\ T(1) & \text{if } n = 1 \end{cases} \quad \text{Recurrence Relation}$$

$$T(n) = 2T(n/2) + n \quad \text{--- (i)}$$

$$T(n/2) = 2T(n/4) + n/2 \quad \text{--- (ii)}$$

$$T(n/4) = 2T(n/8) + n/4 \quad \text{--- (iii)}$$

Sub (iii) in (ii)

$$T(n/2) = 2 \left[2T(n/8) + \frac{n}{4} \right] + \frac{n}{2}$$

$$= 4T(n/8) + \frac{n}{2} + \frac{n}{2}$$

$$T(n/2) = 4T(n/8) + n \quad \text{--- (iv)}$$

Sub (iv) in (i)

for k steps

$$T(n) = 2 \left[4T(n/8) + n \right] + n$$

$$= 8T(n/8) + 2n + n$$

$$= 8T(n/8) + 3n$$

$$= 2^3 T\left(\frac{n}{2^3}\right) + 3n$$

$$2^4 T\left(\frac{n}{2^4}\right) + 4n$$

⋮ for k steps

$$2^k T\left(\frac{n}{2^k}\right) + kn$$

Substitute $T\left(\frac{n}{2^k}\right) = T(1)$

$$\therefore \frac{n}{2^k} = 1$$

$$\therefore 2^k = n = \{4V\}$$

$$\begin{aligned} \text{Apply } \log &= \log 2^k = \log n \\ &= k \log_2 2 = \log n \\ &= k = \underline{\log n} \end{aligned}$$

$$= n \cdot T\left(\frac{n}{n}\right) + \log n \cdot n$$

$$= n \cdot [T(1) + n \cdot \log n]$$

$$= n + n \cdot \log n$$

$$= n(k + \log n)$$

$$O(n) = \boxed{n \cdot \log n}$$

Solve for $\rightarrow \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 5 & 2 & 8 & 3 & 16 & 10 & 20 \end{matrix}$

$$\text{Even } \left\lfloor \frac{0+7}{2} \right\rfloor = 3.5 = 3$$

$$\text{Odd} = \frac{0+8}{2} = \underline{4}$$